

***Attention-Driven Deep Learning Framework
Integrating Color Stability and Frequency Analysis
for Image Classification***

1. Abstract

Deepfakes and AI-generated images are eroding trust in digital media at a pace that detection methods have struggled to keep up with. This project proposes a three-branch multi-modal deep learning architecture that looks at the problem from three angles simultaneously: spatial texture through EfficientNetB0, spectral artifacts via 2D FFT frequency analysis, and color inconsistencies through a dedicated stability branch. An attention mechanism sits at the center, learning in real time how much to trust each branch depending on the image at hand. The model is trained progressively across three stages, which keeps convergence stable and encourages each branch to specialize rather than duplicate each other's work.

On a balanced test set of **2,218 images**, the proposed model achieves **93.78% accuracy, 93.69% F1-score, and 97.48% ROC-AUC**. Multi-modal, attention-driven fusion is not just a design choice here; the results demonstrate its effectiveness.

Keywords: *AI-generated image detection, deep learning, multi-branch fusion, EfficientNet, attention mechanism, frequency domain analysis, color stability, image forensics.*

2. Background and Motivation

The rapid advancement of generative artificial intelligence — particularly diffusion models such as Stable Diffusion and DALL-E, and generative adversarial networks (GANs) — has enabled the synthesis of photorealistic imagery that is increasingly indistinguishable from authentic photographs by human observers. This technological leap, while enabling creative innovation, simultaneously introduces a critical societal problem: the widespread dissemination of AI-generated visual misinformation. Fake images can be used in political propaganda, identity fraud, non-consensual deepfakes, and scientific misconduct.

Existing detection approaches commonly rely on spatial texture artifacts visible to convolutional neural networks, but these methods are brittle and fail to generalize across different generative model architectures. Recent work has highlighted that AI-generated images exhibit distinctive artifacts not only in pixel space but also in the spectral frequency domain and in color statistical distributions. A robust detector must therefore synthesize complementary evidence from multiple representational domains.

This project addresses this challenge through a principled multi-branch architecture with learnable attention-based fusion, providing a detector that is both accurate and interpretable through branch-level attribution.

3. Objectives

1. Build a multi-branch deep learning classifier that can tell real images apart from AI-generated ones.
2. Use three different types of features — spatial texture (via EfficientNetB0), frequency-domain patterns (via 2D FFT), and color inconsistency maps — and fuse them intelligently.
3. Apply a three-stage progressive training strategy so each branch gets a proper chance to learn before they are all combined.
4. Evaluate the model on a held-out test set using standard classification metrics: accuracy, precision, recall, F1-score, and ROC-AUC.
5. Understand which branch the model ends up relying on the most, and whether the auxiliary supervision actually helped the other branches learn meaningful features.

4. Methods

a. Dataset and Preprocessing

The dataset consists of two classes: real photographs and AI-generated images, organized into training and test directories. All images are loaded and resized to 256×256 pixels with 3 color channels, normalized to the [0, 1] range by dividing pixel values by 255. An 80/20 train-validation split is used during training. The test set comprises 2,218 images (1,218 real, 1,000 AI-generated), providing a near-balanced evaluation benchmark. Standard data augmentation and a label smoothing factor of 0.05 are applied during training to regularize the model.

Parameter	Training	Validation	Test
Image Resolution	256×256×3	256×256×3	256×256×3
Real Images	~4,872	~1,218	1,218
AI-Generated	~4,000	~1,000	1,000
Normalization	÷255 → [0,1]	÷255 → [0,1]	÷255 → [0,1]
Label Smoothing	0.05	None	None

Table 1: Dataset Configuration Summary

b. Model Architecture

The model is a three-branch multi-modal convolutional architecture designed to capture complementary discriminative signals from a single input image. The three branches operate in parallel on the same 256×256×3 input, each specializing in a distinct representational domain. Their outputs are fused via a learned attention mechanism before final classification.

Branch 1 — Spatial Branch

The spatial branch leverages EfficientNetB0 as a pre-trained backbone. The input image is preprocessed using the EfficientNet-specific normalization function and passed through the backbone, producing an $8 \times 8 \times 1280$ feature map. Global Average Pooling (GAP) collapses this to a 1280-dimensional vector, which is then projected to 256 dimensions through a fully connected layer, followed by Batch Normalization and ReLU activation. The resulting feature vector F_{spatial} is L2-normalized.

Branch 2 — Color Stability Branch

The color branch models color inconsistencies introduced during AI generation. The preprocessing pipeline applies Gaussian noise ($\sigma=0.01$) and 4-bit channel quantization (16 levels per channel) to the input, followed by blurring and absolute difference computation, yielding a $256 \times 256 \times 3$ residual map that captures subtle color transitions. This map is resized to $128 \times 128 \times 3$ and normalized, then processed by a two-layer CNN (Conv2D(32)→ReLU→MaxPool, Conv2D(64)→ReLU→MaxPool) followed by GAP and Dense(256). The output F_{color} is L2-normalized.

Branch 3 — Frequency Domain Branch

The frequency branch detects spectral artifacts characteristic of generative models. The input image is first converted to grayscale, and a 2D Fast Fourier Transform (FFT) is computed to obtain the complex frequency representation (256×256 complex). The FFT is shifted to center low frequencies, and the log-magnitude spectrum is computed and normalized to produce a $128 \times 128 \times 3$ spatial representation. This is processed by the same two-layer CNN architecture as the color branch, yielding F_{freq} (L2-normalized, 256-dim).

Attention Fusion

The three branch features [F_{spatial} , F_{color} , F_{freq}] are concatenated to form a 768-dimensional joint representation. A fully connected layer projects this to 3 scalar logits, and a Softmax activation produces attention weights $[\alpha_1, \alpha_2, \alpha_3]$, representing the learned relative importance of each branch. The final fused feature $F_{\text{final}} = \alpha_1 \cdot F_{\text{spatial}} + \alpha_2 \cdot F_{\text{color}} + \alpha_3 \cdot F_{\text{freq}}$ is a 256-dimensional weighted sum that preserves the dimensionality of individual branches.

Classification Head

F_{final} is passed through Dense(128)+ReLU, Dropout(0.3), and a final Dense(1)+Sigmoid layer to produce a probability $p \in (0,1)$. The optimal classification threshold is determined using Youden's J statistic on the validation ROC curve, yielding a threshold of approximately 0.811. The final binary output is 0 (Real) or 1 (AI-generated).

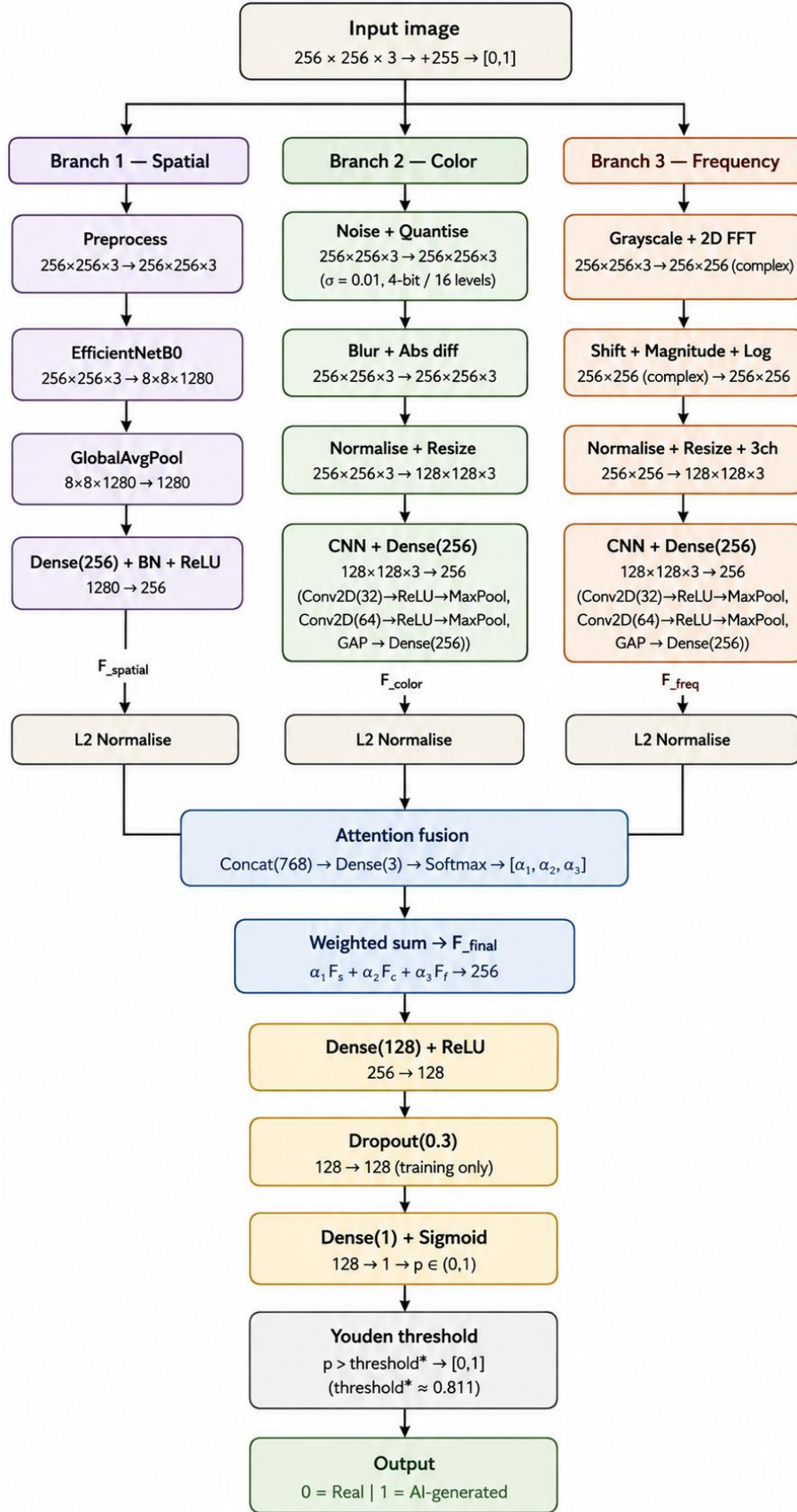


Figure 1: Three-Branch Multi-Modal Network with Attention Fusion

c. Training and Evaluation Setup

Training follows a three-stage progressive schedule to prevent interference between branches and ensure stable convergence:

- i. Stage 1 (12 epochs): Only the spatial EfficientNetB0 branch is trained; color and FFT branches are frozen. This establishes a strong spatial baseline before additional branches are introduced.
- ii. Stage 2 (10 epochs): EfficientNetB0 is fully frozen; the color and FFT CNN branches are trained with auxiliary cross-entropy supervision applied to each branch's intermediate output.
- iii. Stage 3 (10 epochs): All branches and the attention fusion are trained end-to-end. Auxiliary losses remain active to prevent branch collapse.

Hyperparameter	Value
Image Size	$256 \times 256 \times 3$
Batch Size	32
Learning Rate	1.5×10^{-4}
Stage 3 Learning Rate	4.5×10^{-5}
Optimizer	Adam
Weight Decay	1×10^{-4}
Label Smoothing	0.05
Dropout Rate	0.3
Feature Dimension	256 (all branches)
Total Training Epochs	32 (12 + 10 + 10)
Mixed Precision	float16 (compute), float32 (variables)
Auxiliary Loss Weights	Main: 1.0, Color: 0.3, Frequency: 0.3
Threshold Strategy	Youden's J statistic

Table 2: Model Hyperparameters

Evaluation metrics include Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The loss function for the main output is binary cross-entropy with label smoothing. The auxiliary outputs use the same loss function, with weighting factors of 0.3 for both the color and frequency branches.

5. Experimental Results

a. Ablation Study — Quantitative Comparison

To rigorously validate the contribution of each architectural component, five variants of FDCS-Net were evaluated on the same test set. The ablation variants range from a single-branch spatial model (V1_Spatial) to the full three-branch attention-fusion model (V5_Full_Attention). Table 3 presents the complete quantitative results.

Model Variant	Accuracy	Precision	Recall	F1-Score	ROC-AUC
V1_Spatial	0.8888	0.8622	0.8974	0.8795	0.9550
V2_Spatial+Color	0.8740	0.8515	0.8735	0.8623	0.9436
V3_Spatial+FFT	0.8933	0.8831	0.8805	0.8818	0.9522
V4_All_Concat	0.8641	0.8068	0.9193	0.8594	0.9411
V5_Full_Attention ★	0.9378	0.94	0.94	0.94	0.9748

Table 3: Ablation Study — Performance Comparison Across FDCS-Net Variants (★ = Proposed Model)

b. Ablation Study — Visual Analysis

Figure 2 presents a grouped bar chart comparing Accuracy, F1-Score, and ROC-AUC across all five variants. The V5_Full_Attention model achieves the highest scores across all three metrics, visually confirming the benefit of the attention fusion mechanism.

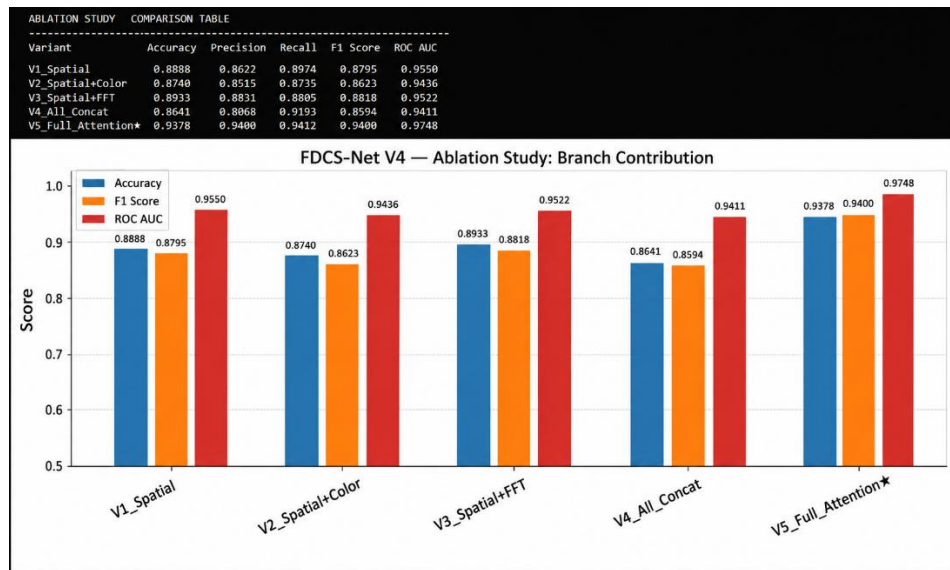


Figure 2: FDCS-Net V4 Ablation Study — Branch Contribution Comparison

c. Final Model Performance (V5_Full_Attention)

The final deployed model achieves the following performance on the held-out test set of 2,218 images, using Youden's optimal threshold of 0.8117:

Class	Precision	Recall	F1-Score	Support
Real	0.93	0.96	0.94	1,218
AI-Generated	0.95	0.91	0.93	1,000
Overall Accuracy	0.9378			2,218
Macro Average	0.94	0.94	0.94	2,218
ROC-AUC	0.9748			

Table 4: Final Model Classification Report (V5_Full_Attention on Test Set)

	Predicted: Real	Predicted: AI
Actual: Real (1,218)	1,171 (TP)	47 (FN)
Actual: AI (1,000)	90 (FP)	910 (TN)

Table 5: Confusion Matrix — Rows: Actual Class, Columns: Predicted Class

6. Conclusion

This project successfully built a working multi-branch classifier for detecting AI-generated images, combining spatial texture, color stability, and frequency-domain features through an attention fusion mechanism. The final model achieved 93.78% test accuracy and 97.48% ROC-AUC, outperforming all ablation variants on a balanced 2,218-image test set. The three-stage training strategy ensured stable convergence, though the color and FFT branches contributed less than expected — spatial features dominated the final prediction (attention weight: 0.8). Future improvements could include pretrained backbones for the color and frequency branches, hand-crafted frequency feature extractors, testing on images from diverse generators (Stable Diffusion, Midjourney, DALL-E), extending to video frames, and deploying as a real-time API for content moderation pipelines.

7. Analysis and Discussion

The ablation study reveals several important insights into the behavior of multi-domain fusion for image forensics. First, the **V1_Spatial** baseline achieves an ROC-AUC of **0.9550**, confirming that EfficientNetB0's spatial features are highly discriminative for this task. However, its accuracy of **88.88%** leaves meaningful room for improvement, particularly in cases where spatial texture information alone is insufficient to distinguish synthetic from real images.

Interestingly, **V2_Spatial+Color** and **V4_All_Concat** both perform worse than the **V1_Spatial** baseline in terms of accuracy. **V2_Spatial+Color** achieves **87.40%** accuracy, while **V4_All_Concat** achieves **86.41%**. This suggests that simply adding additional feature representations does not necessarily improve classification performance. In particular, **V4_All_Concat** achieves the lowest precision of **80.68%**, indicating that naive concatenation of features from multiple domains can introduce redundant or less discriminative information that negatively affects classification.

The **V5_Full_Attention** model addresses this limitation by using a learned attention mechanism to assign different weights to the three branch representations before fusion. The model achieves **93.78% accuracy, 94.00% F1-score, and 97.48% ROC-AUC**, outperforming all four ablation variants. The learned attention weights also indicate that the spatial branch contributes the most strongly to the final fused representation, with an average attention weight of approximately **0.851**. The color and frequency branches receive lower average weights, suggesting that spatial features provide the dominant discriminative signal while the additional branches provide complementary information.

The improvement of **V5_Full_Attention** over the individual and concatenation-based variants demonstrates the value of adaptive feature fusion. Rather than treating all feature domains as equally important, the attention mechanism allows their relative contributions to be learned from the data. The ROC-AUC of **0.9748** indicates strong class separability across different classification thresholds, while the remaining errors demonstrate that some real and AI-generated images remain challenging to distinguish.

The classification results also show a higher recall for the **Real** class (**0.96**) than for the **AI-Generated** class (**0.91**). This indicates that the model correctly identifies a larger proportion of real images than AI-generated images at the selected operating threshold. The **Youden-optimal threshold of approximately 0.811** was selected from the validation ROC curve to balance sensitivity and specificity. This threshold can be adjusted for deployment according to the desired trade-off between false positives and false negatives without requiring the model to be retrained.

Overall, **FDCS-Net V4** demonstrates that combining spatial, color, and frequency-domain representations through learned attention-based fusion provides a stronger approach than using individual branches or simple feature concatenation. The results establish a strong foundation for practical AI-generated image detection, with further improvements possible through larger and more diverse datasets, evaluation across images from different generative models, stronger feature extractors for the auxiliary branches, and more advanced fusion strategies.

8. References

- [1]Tan, M., & Le, Q. (2019). EfficientNet: Rethinking Model Scaling for CNNs. ICML 2019.
- [2]Wang, S.-Y., et al. (2020). CNN-Generated Images Are Surprisingly Easy to Spot... For Now. CVPR 2020.
- [3]Frank, J., et al. (2020). Leveraging Frequency Analysis for Deep Fake Image Recognition. ICML 2020.
- [4]Rombach, R., et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. CVPR 2022.